

Clustered, Longitudinal and Split-Plot Data

Most data arrive in groups. Plots lie in fields, pupils sit in classrooms, patients return for repeated visits, firms are followed for twenty years. Two observations from the same group are more alike than two from different groups, and the error covariance is no longer $\sigma^2\mathbf{I}$. This chapter is the applied half of Part VII: it takes the general Gauss–Markov theory of [Chapter 31](#) and the mixed models of [Chapter 32](#) and puts them to work on the three structures that account for most correlated data in practice. A split-plot experiment randomizes at two levels and therefore has two error terms, so some comparisons are precise and others are not. A repeated-measures analysis needs a condition on the covariance that is weaker than equicorrelation and stronger than nothing, and there are three standard ways of dealing with its failure. A longitudinal study needs a covariance model chosen from a small catalogue, checked against the data, and hedged with a standard error that survives its being wrong. The chapter closes with growth curves, where each subject has its own trajectory and the whole point is to borrow strength across subjects.

What you need. [Chapter 6](#) (projection in a general inner product, Definition 6.9.1, and when it agrees with least squares, Theorem 6.9.3), [Chapter 9](#) (expected mean squares, Theorem 9.6.1), [Chapter 16](#) (balanced layouts as Kronecker products, Lemma 16.1.1, Theorem 16.3.1), [Chapter 18](#) (randomized blocks, Theorem 18.4.1), [Chapter 19](#) (the effect of a wrong covariance, Theorem 19.2.2), [Chapter 21](#) (sandwich estimators, Theorem 21.4.2, autoregressive errors, Definition 21.6.1), [Chapter 29](#) (Definition 29.2.2), and the two chapters before this one: generalized least squares and its inference (Definition 31.1.1, Theorem 31.1.1, Theorem 31.2.2) and the mixed model with its predictors and likelihoods (Definition 32.2.1, Theorem 32.3.1, Theorem 32.3.2, Definition 32.4.1). The distribution theory is that of Part I, used directly and not only through those chapters: the noncentral F , the mean and variance of a quadratic form and the independence of orthogonal projections from [Chapter 4](#) (Definition 4.1.2, Theorem 4.2.3, Theorem 4.4.4); whitening, the covariance of a linear function and the mean of a quadratic form from [Chapter 2](#) (Proposition 2.2.4, Theorem 2.2.1, Theorem 2.3.1); the Woodbury identity and a matrix square root from [Chapter 1](#) (Theorem 1.6.5, Theorem 1.7.6); Aitken’s corollary from [Chapter 7](#) (Corollary 7.2.3); and, in [Section 33.3](#), the centred Wishart lemma of [Chapter 14](#) (Lemma 14.2.3).

33.-1.1 Roadmap

- [Split-plot designs](#)
- [Clustered data and compound symmetry](#)
- [Repeated measures](#)
- [Covariance models for longitudinal data](#)
- [Growth curves](#)
- [Summary and notes](#)